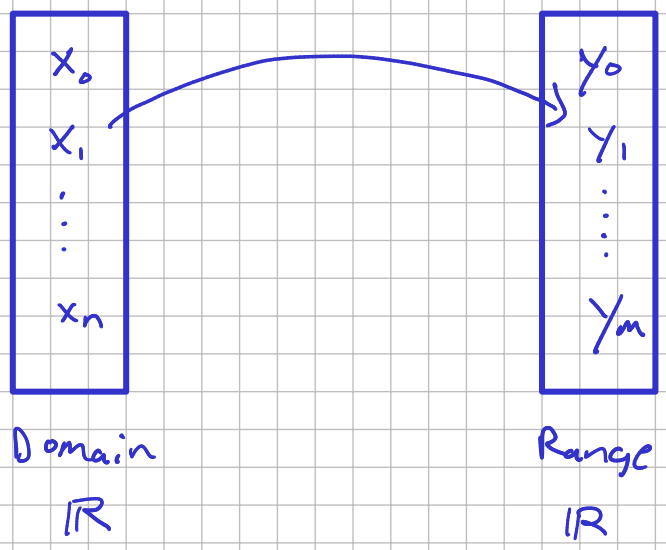
Review - Concepts and Strategies

Concept

1) <u>Function</u>: a set of inputs (domain), outputs (range), and a set of rules for <u>mapping</u> each input to exactly one output.

$$x \longrightarrow \boxed{f(x)} \longrightarrow y$$



$f: \mathbb{R} \to \mathbb{R}$

Example: $f(x) = x^2$

Domain $\{x \mid x \in \mathbb{R}\}$

Range $\{y \mid y \geq 0\}$

Strategy

1a) <u>Domain</u>

- Consider x expr. in isolation
- Note any restrictions on x
- Express x in terms of restriction
- solve for x

1b) <u>Range</u>

- Solve $y = f(x)$ for x
- consider y-expr. in isolation
- Observe any restrictions on y
- Express y expr. in terms of restr.
- Solve for y

1c) <u>Zeros</u>

- Set $f(x) = 0$
- Solve for x

1d) <u>Y-Intercepts</u>

- Evaluate $y = f(0)$
- Intercept $(0, f(0))$

Concept

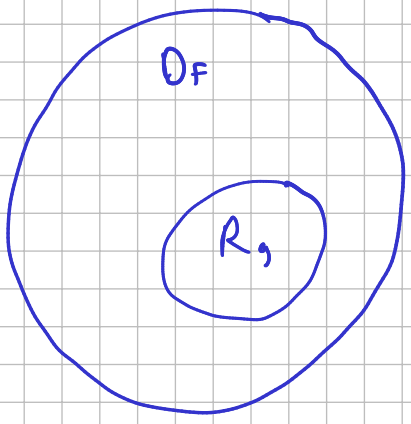
2) <u>Composite Function</u>: a func. that has another func. as its input.

$$f \circ g(x) = f(g(x))$$

$f(x)$ has domain D_f
has range R_f

$g(x)$ has domain D_g
has range R_g

Rule: If $R_g \subseteq D_f$, then $f \circ g(x)$ has domain D_g



$R_g \subseteq D_f$

Strategy

2a) <u>Domain</u>

- Find D_f
- Find R_g
- check $R_g \subseteq D_f$
- If True:

$$D_{f \circ g} = D_g$$

- If False:
  - Find $D_{f \circ g}$
  - From $f(g(x))$

Concept

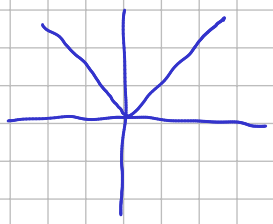
3) Absolute Value Function:

$$f(x) = |x|$$

$$= \begin{cases} -x, & x < 0 \\ x, & x > 0 \end{cases}$$

Example: $f(x) = 2|x-3|$

Graph of $f(x) = |x|$



Strategy

3a) <u>Solving for x:</u>

- Isolate expr. with '| |' on one side
- Consider epr. $|A| = B$
- If $B < 0$, there is no solv
- Split into 2 cases:

  $A = B$ and $A = -B$
- Solve each for x

Concept	Strategy

4) Polynomials:

functions of the form:

$a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$

Degree: n

Leading coefficient: a_n

If n is even $(2, 4, \cdots)$

$a_n > 0$: $f(x) \to \infty$ as $x \to \pm\infty$

$a_n < 0$: $f(x) \to -\infty$ as $x \to \pm\infty$

If n is odd $(1, 3, 5 \cdots)$

$a_n > 0$: $f(x) \to \infty$ as $x \to \infty$
$f(x) \to -\infty$ as $x \to -\infty$

$a_n < 0$: $f(x) \to -\infty$ as $x \to \infty$
$f(x) \to \infty$ as $x \to -\infty$

4a) Finding zeros (quadratic)

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

- If $b^2 - 4ac > 0 \Rightarrow$ 2 solutions
- If $b^2 - 4ac < 0 \Rightarrow$ No real solutions
- If $b^2 - 4ac = 0 \Rightarrow$ 1 solution

4b) Axis of Symmetry (Quadratic)

$$\frac{z_1 + z_2}{2} = \frac{-b}{2a} = x$$

where z_1 and z_2 are zeros!

4c) vertex (min/max of a quadratic)

$$\rightarrow \left[\frac{-b}{2a}, f\left(\frac{-b}{2a}\right) \right]$$

4d) Finding zeros (Cubic)

- Factor an x out
- use quadratic

Example: $2x^3 + x^2 + 2x + 4$

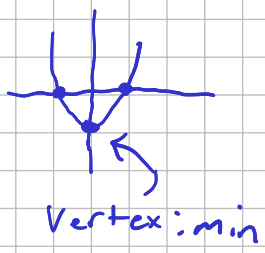
$\Rightarrow x(2x^2 + x + 2) + 4$

solve for zeros

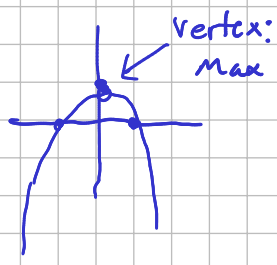
Concept

Quadratic $n = 2$

$a_n > 0$:

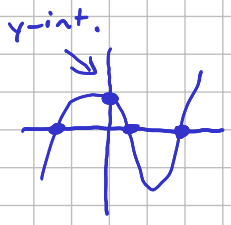


$a_n < 0$:

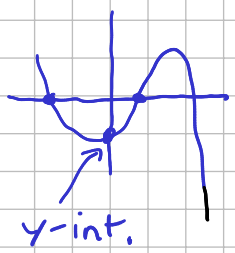


Cubic $n = 3$

$a_n > 0$:
left tail ↓



$a_n < 0$:
left tail ↑



For <u>any</u> polynomial of deg. n:

n is even: ↑↑ ↓↓

n is odd: ↑↓ ↓↑

Strategy

4e) **Graphing (Quadratic)**

- Find zero
- Find the axis of symmetry
- Find the vertex
- Use sgn(a_n) for direction

4f) **Graphing (Cubic)**

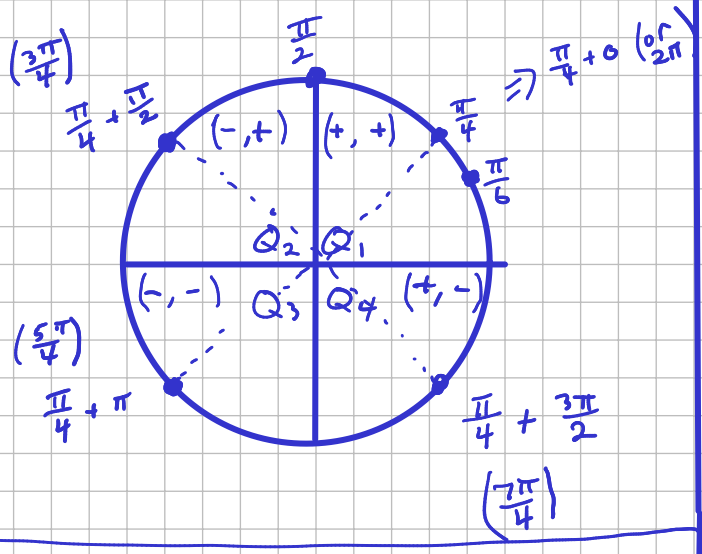
- Find zeros
- Find y-intercept
- Use a_n for direction of tails

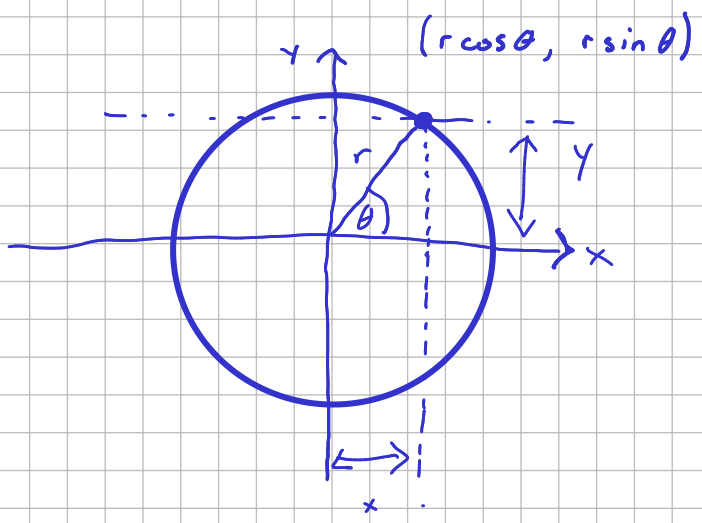
Concept

5) <u>Trigonometric Functions</u> describe cyclic or oscillating behavior.

→ Periodic!

- We measure angles in <u>radians</u>





$\cos\theta = \frac{x}{r}$ $\qquad$ $\sin\theta = \frac{y}{r}$

θ	$\cos\theta$	$\sin\theta$
0	1	0
$\pi/6$	$\sqrt{3}/2$	$1/2$
$\pi/4$	$\sqrt{2}/2$	$\sqrt{2}/2$
$\pi/3$	$1/2$	$\sqrt{3}/2$
$\pi/2$	0	1

Strategy

5a) <u>Finding Corresponding Angles</u>

- Start in Quadrant Q
- Find Target Quadrant Q_T
- Calculate $(Q_T - Q)\frac{\pi}{2}$

5b) <u>Find (x, y)</u>

- Find angle corresponding to θ
- Find $(r\cos\theta,\ r\sin\theta)$
- Apply sign (+/−) for Q_T

Concept

6) Properties of Functions

Injectivity (one-to-one):

$$\boxed{\text{If } f(x_1) = f(x_2) \text{ then } x_1 = x_2}$$

Surjectivity (onto):

- Depends on a target set Y_T (codomain)
- For function $f(x)$ with range R_f:

$$\boxed{\forall y \in Y_T : y \in R_f}$$

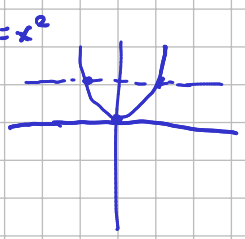
→ If every y in the target set Y_T exists in the range of the given function R_f, then $f(x)$ is surjective!

Bijectivity:

Both injective & surjective!

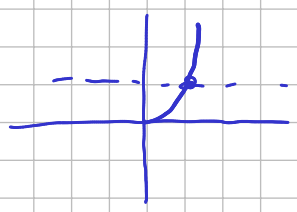
Restricting domain D_f to enforce injectivity of $f(x)$

$f(x) = x^2$



Domain ($x \in \mathbb{R}$)

$f(-2) = 4 = f(2)$



Domain ($x \geq 0$)

$\boxed{\rightarrow \text{Injective}}$

Lecture Notes: 2_Functions, 3_F...

Strategy

6a) Injectivity

- Check mathematical definition condition
- Horizontal line taps graph at most once

6b) Surjectivity

- Given $f(x)$, Y_T
- Find R_f (strategy 1b)
- Verify $\forall y \in Y_T : y \in R_f$
- Graph has no gaps along y-axis over Y_T

6c) Bijectivity

- Check 6a and 6b
- Graph one tap, no gap!

"touches once, no gaps along Y_T"

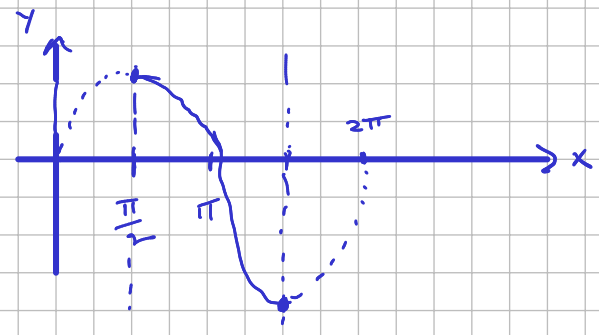
6d) Finding a restricted domain to enforce injectivity.

- Sketch a graph
- Mark the endpoints

Example

$f(x) = \sin x$

Find a restricted domain over which $\sin x$ is injective.



1. Sketch
2. Mark endpoints
3. Write interval → $\left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$

Concept	Strategy

7) Symmetry and Translation

Translations: Horizontal

$f(x-a)$ Translates $f(x)$ right by a units

$f(x+a)$ Translates $f(x)$ left by a units

Translations: Vertical

$f(x)+b$ Translates $f(x)$ up by b units

$f(x)-b$ Translates $f(x)$ down by b units

Reflections:

$\boxed{-f(x)}$ is a reflection of $f(x)$ about the x-axis

$\boxed{f(-x)}$ is a reflection of $f(x)$ about the y-axis

→ Axis Symmetry (y-axis)

If $\boxed{f(x) = f(-x)}$ then:

$f(x)$ is even and symmetric about the y-axis.

→ Point Symmetry:

If $\boxed{f(-x) = -f(x)}$ then

$f(x)$ is odd and symmetric about the origin.

7a) Change the origin of a function by translating it to point (a, b)

a: horizontal
b: vertical

7b) Reflect a function:

- use $-f(x)$ for the x-axis
- use $f(-x)$ for the y-axis

7c) Finding whether a function is symmetric about $x = a$

- Remove horizontal translation
→ set $a = 0$
- Test $f(x) = f(-x)$

7d) Finding whether a function is symmetric about a point (a, b)

- Remove translations $a = b = 0$
- Test $f(-x) = -f(x)$

Example: $f(x) = -(x+1)^3 + 4$

is $f(x)$ symmetric about $(-1, 4)$?

Intuition: $-x^3$

$f(-x) = x^3$ ✓

$-f(x) = x^3$

YES → symmetric about $(-1, 4)$

Concept	Strategy

8) Inverse Functions

If a function is injective, it has an inverse

$$f^{-1}(x)$$

8a) Finding the Inverse of $f(x)$

- verify injectivity
- solve $y = f(x)$ for x
- swap x and y
- write $y = f^{-1}(x)$

9) Exponential Functions: have the form

$$b^x$$

$\boxed{\text{Special case: } e^x}$

$b > 1$: increases $(-\infty, \infty)$

$0 < b < 1$ Decreases $(-\infty, \infty)$

9a) Solving an exp. Func. for x

- Take the log (inverse func) of both sides

$$b^x = c$$

$$= \log_b(b^x) = \log_b(c)$$

$$= x = \log_b(c)$$

10) Logarithmic Functions: are the inverse of exponential functions.

$$f(x) = \log_b(x)$$

$\boxed{\text{Special case: } \ln x}$

10a) Solving log. funcs for x

- Raise both sides to a power of b

$$\log_b(x) = c$$

$$b^{\log_b(x)} = b^c$$

$$x = b^c$$

Concept

11) Limits

- Let $f(x)$ be defined over an open interval around a, possibly including a.
- If all values for $f(x)$ approach a single, real number L as $x \to a$ $(x \neq a)$ then:

$$\boxed{\lim_{x \to a} f(x) = L}$$

True if (and only if):

$$\boxed{\lim_{x \to a^-} f(x) = L \quad \text{and} \quad \lim_{x \to a^+} f(x) = L}$$

Otherwise

$\lim_{x \to a} f(x)$ DNE!

Strategy

11a) Continuity of $f(x)$ at $x = a$

For $f(x)$ to be continuous at a:

1. $f(a)$ is defined
2. $\lim_{x \to a} f(x)$ exists
3. $\lim_{x \to a} f(x) = f(a)$

11b) Continuity of $f(x)$ over $I = [a, b]$

Check 3 conditions:

1. $f(x)$ is continuous for all $x \in (a, b)$
2. $\lim_{x \to a^+} f(x) = f(a)$
3. $\lim_{x \to b^-} f(x) = f(b)$

11c) Types of Discontinuities

1) Removable "Hole" Discontinuity

$\lim_{x \to a} f(x)$ exists but $\neq f(a)$

Fix: Redefine as piecewise with $f(a) = L$

Example: $f(x) = \dfrac{x^2 - 4}{x - 2} = \begin{cases} \frac{x^2-4}{x-2}, & x \neq 2 \\ 4, & x = 2 \end{cases}$

2) Jump or Infinite Discontinuity

- left/right limits not equal or infinite

$$\boxed{\begin{gathered} \lim_{x \to a^-} f(x) \neq \lim_{x \to a^+} f(x) \\ \text{OR} \\ \lim_{x \to a} f(x) = \pm\infty \end{gathered}}$$

Concept

12) Derivative of $f(x)$

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

Power Rule

$$f(x) = x^n$$
$$f'(x) = n x^{n-1}$$

Product Rule

$$h(x) = f(x) g(x)$$
$$h'(x) = f'(x) g(x) + g'(x) f(x)$$

Quotient Rule

$$h(x) = \frac{f(x)}{g(x)}$$
$$h'(x) = \frac{f'(x) g(x) - g'(x) f(x)}{g(x)^2}$$

Chain Rule

$$h(x) = f \circ g(x) = f(g(x))$$
$$h'(x) = f'(g(x)) g'(x)$$

Strategy

12a) Chain Rule

1. Identify $f(x)$ and $g(x)$
2. Find $f'(x)$, then $f'(g(x))$
3. Find $g'(x)$
4. $h'(x) = f'(g(x)) g'(x)$

12b) Differentiability of $f(x)$

1. Identify possible problem points:
   - Denominators: $f(x)/g(x)$
   - Even roots: $\sqrt{x}$
   - Absolute values: $|x|$ (corners)
   - Piecewise: boundary points
   - Non-polynomials: $\ln(x)$ (cont. for $x > 0$)
2. Verify $f(x)$ is continuous at $x = a$
3. Verify left/right derivatives exist and are equal:

$$\lim_{h \to 0^-} \frac{f(a+h) - f(a)}{h} = \lim_{h \to 0^+} \frac{f(a+h) - f(a)}{h}$$

Concept

13) Higher-Order Derivatives

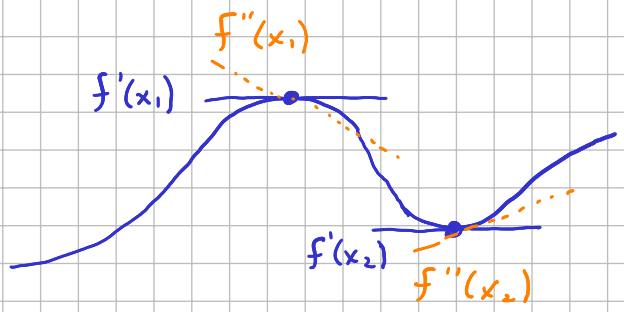
$f'(x) = \frac{df}{dx}$ 1st-order

$f''(x) = \frac{d^2f}{dx^2}$ 2nd-order

$f'''(x) = \frac{d^3f}{dx^3}$ 3rd-order

$\vdots$

$f^{(n)}(x) = \frac{d^nf}{dx^n}$ nth-order



Monotonicity

- Strictly Monotonically increasing
  $f(x_2) > f(x_1)$
  (Increasing)

- Monotonically increasing
  $f(x_2) \geq f(x_1)$
  (Not decreasing)

- same for decreasing

Strategy

13a) Local minima/maxima

1. Find $f'(x)$
2. Find critical points: $f'(x) = 0$
3. Take second derivative and evaluate at each critical point x_0:

$f''(x_0) > 0 \Rightarrow$ local min

$f''(x_0) < 0 \Rightarrow$ local max

13) Monotonicity (using 1st-order)

1. Find $f'(x)$
2. Determine sign of $f'(x)$ on intervals between critical points by choosing an arbitrary point:
   $x \in (a, b)$

$f'(x) > 0 \quad \forall x \in (a, b)$:

Strictly increasing on (a, b)

$f'(x) < 0 \quad \forall x \in (a, b)$:

Strictly decreasing on (a, b)

* If domain is restricted, check end points separately to close interval:
[a or (a